

Product Requirements Document

Solving Order Drop — Reducing Post-Payment Cancellations via Pre-Checkout Availability Gating

1. Executive Summary

FoodSwift experienced a meaningful decline in key engagement metrics between January and March — orders per month dropped from 3.2 to 2.8, and NPS scores fell in the same window. This document diagnoses the root cause and proposes a two-sided product solution to reverse the trend.

The core insight is that users are not uninstalling — they are ordering less frequently. This 'silent churn' pattern is driven by a single friction point: post-payment order cancellations caused by restaurant kitchen unavailability. Cancellations that occur after payment are disproportionately damaging to trust because users have committed intent and money, only to be failed at the last step.

Metric	Jan Baseline	Post-March	Delta
Orders / Month	3.2	2.8	-12.5%
NPS	Stable baseline	Observed decline	Negative
Post-Cancel Re-order	—	Low	Concerning

Solution summary: Implement (a) ML-powered restaurant reliability ratings visible pre-checkout, and (b) a 90-second kitchen capacity confirmation gate that fires before payment is processed. When a kitchen declines, smart rerouting to "Highly Reliable" alternatives preserves the user's order intent.

2. Problem Statement

2.1 Observed Metric Deterioration

Between January and March, FoodSwift's two primary health metrics — orders per user per month and Net Promoter Score — declined measurably. The drop in ordering frequency (3.2 → 2.8) is particularly telling because it reflects a change in **willingness to trust the platform**, not a change in user need or appetite for the product category.

2.2 Why 'Silent Churn' is the Right Frame

Traditional churn is tracked via uninstalls or account deletion. Silent churn — reduced usage frequency without leaving — is harder to detect but equally damaging to long-term GMV. A user who orders 2.8x instead of 3.2x per month represents a 12.5% revenue reduction with zero visible signal in retention dashboards.

- Users do not uninstall after a bad experience; they reduce ordering cadence
- Lower NPS from failed orders means fewer referrals and organic acquisition
- Frequency decline compounds: a user who skips one week often re-baselines at a lower cadence
- Discounts or coupons cannot restore trust broken by operational failures

2.3 The Central Problem

"Cancellations don't hurt because orders fail — they hurt because they fail late. A failure before checkout is a feature gap. A failure after payment is a betrayal."

3. Target User Persona

Habitual Urban Food Purchaser — Primary Persona

Demographic	Behavioral	Sensitivity
Age 20–35 Student / early-career professional Tier-1 cities Bachelor or newly married	2–4 orders/month AOV ~\$450 (solo or light 2-person) Convenience-first High fulfillment expectation	Values reliability > discounts Does NOT uninstall after failures Reduces ordering frequency instead NPS drops, referrals dry up

3.1 Key Behavioral Insight

This persona does not make a deliberate decision to stop using FoodSwift. The reduction in ordering frequency is subconscious — a learned distrust that manifests as choosing to cook, grab something on the way home, or try a competitor on days when the usual ordering impulse arises. This makes recovery time-sensitive: the longer the low-frequency pattern persists, the harder it is to restore.

4. Root Cause Analysis — Hypotheses

Three seasonal and behavioral hypotheses explain the Jan–March deterioration pattern. These are not mutually exclusive — all three likely compound the observed decline.

H1 — January Stability Baseline

Why	December bonuses and post-holiday routine re-entry create unusually stable, predictable demand in January. Users are willing to spend on sides and higher-priced restaurants. Weekday ordering is consistent. This makes January a reliable control period for metrics.
Effect	Higher AOV, consistent weekday ordering, stable order count. Metrics appear healthy.

H2 — February: Basket Shrinkage from Necessity-Driven Orders

Why	Fewer days in February, fewer weekend gatherings, and post-January financial tightness push users toward quick, solo meals rather than planned or group orders. Restaurants promote budget combos. Users subconsciously reduce basket size — not order frequency.
Effect	Lower AOV (not lower order count). Restaurants observe smaller basket sizes and fewer add-ons.

H3 — March: Demand Spike + Kitchen Capacity Mismatch

Why	Fiscal year-end work stress and exam season simultaneously spike demand at non-standard hours. Restaurants fail to update item availability dynamically, and staffing constraints limit capacity. Orders that arrive during peak stress moments (exams, deadlines) get cancelled post-payment — exactly when the emotional cost to users is highest.
Effect	Post-March NPS decline and frequency drop. Late cancellations during high-need moments cause the most trust damage.

5. Solution Design

The solution addresses the problem at two levels: the customer experience layer and the restaurant operations layer. Together they shift risk detection before the payment moment.

5.1 Guiding Design Principles

- **Shift risk detection before checkout, not after payment.** The failure window should close before money changes hands.
- **Alignment through confirmation gating.** A restaurant should explicitly accept an order before it is confirmed to the user.
- **Fast, transparent rerouting over silent failures.** When a kitchen cannot deliver, provide an immediate alternative rather than an error message.

5.2 Feature 1 — ML Restaurant Reliability Ratings (Customer-Side)

Each restaurant is scored based on its last 10 order completion outcomes. This score updates dynamically as new orders resolve.

Completion Rate	Label Displayed	UX Treatment
≥ 8/10 (80%) orders completed	■ Highly Reliable	Green badge on restaurant card
< 8/10 threshold	■ May Face Delays	Amber warning + expectation-setting copy

5.3 Feature 2 — 90-Second Kitchen Confirmation Gate (Restaurant-Side)

When a user proceeds to checkout, the order is held in a pending state and simultaneously sent to the kitchen dashboard. The restaurant has 90 seconds to confirm that it has sufficient capacity — covering chef availability, ingredient inventory, and delivery slot availability.

- **Accept within 90s:** Order proceeds to payment and confirmation. User sees 'Kitchen is preparing your food.'
- **Decline within 90s:** No payment is charged. User is immediately shown 'Highly Reliable' alternatives offering the same cuisine, factoring in cuisine-specific ratings.
- **No response within 90s:** Treated as a soft decline. System surfaces alternatives proactively.

5.4 Order Flow Comparison

Step	Old Flow (Broken)	New Flow (FoodSwift Solution)
1	Browse restaurant	Browse restaurant (sees reliability badge)
2	Select items & pay	Select items → Pre-checkout kitchen check triggered
3	Order sent to kitchen	Kitchen accepts/declines within 90s (before payment)
4	Kitchen rejects after payment	If declined: instant reroute to alternatives

5	User gets generic error	User completes order → trust maintained
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6. Success Metrics & Measurement

6.1 Primary Metrics (First-Order Benefits)

These metrics directly measure whether the core problem — eroded user trust from post-payment cancellations — has been resolved.

Metric	Baseline	Target	Rationale
Orders / User / Month	2.8	3.2 (+14.3%)	Restore pre-drop frequency
User NPS	Post-March low	+8–12 pts	Reverse trust erosion from cancellations
Post-Cancellation Re-order Rate	Low	+20% lift	Measure reroute effectiveness
Order Completion Rate	—	> 95%	Track successful first-attempt order

6.2 Secondary Metrics (Second-Order Benefits)

These leading indicators confirm the solution is working at the operational level and will drive the primary metrics over time.

- **Share of orders from 'Highly Reliable' restaurants:** Should increase as users trust and prefer reliable restaurants.
 - **% of orders confirmed in first 90 seconds:** High confirmation rates indicate healthy restaurant capacity management.
 - **Number of users reordering after an early (pre-payment) cancellation:** Validates rerouting UX effectiveness.
 - **Distribution shift in restaurant reliability scores:** Restaurants should improve completion rates to avoid 'May Face Delays' labels.
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